The structure of the proof

An orientation preserving C^3 local universal deformation
of a homogeneous incompressible isotropic hyperelastic solid

The constant principal stretch case
Three pairwise distinct principal stretches; suppose $h_1 h_2 h_3 \neq 0$ on an open set.

Universal equilibrium and compatibility
Finite equations for the principal frame and its derivatives (Sections 2–3)

Weighted normalization gives the state map X on a dense chart.
$\rho = \operatorname{rank} d\mathsf{X} \in \{0, 1, 2, 3\}$; the common similarity algebra has dimension $3 - \rho$.

Ranks zero and one
At least two common similarities
Similarity algebras and the frame equations
Exclusion (Section 4)

Rank two
One common similarity
Isometries or dilations; continuation and cylindrical limits
Exclusion (Sections 5–7)

Rank three
Projection onto the two abnormality ratios
Algebraic fibre estimates give $\dim Y \leq 2$
Exclusion (Section 8)

$h_1 h_2 h_3 \equiv 0$; analyticity yields a fixed label with $h_i \equiv 0$.
The plane field orthogonal to that principal direction is integrable.

Kafadar's theorem
Homogeneous and logarithmic cylindrical deformations in the constant distinct case

Earlier classifications
Variable principal invariants or repeated principal values

Analytic continuation through critical points of the principal invariant map
and removable coordinate singularities (Section 9)

The six classical families $\mathcal{F}_0, \ldots, \mathcal{F}_5$
Local completeness for orientation preserving C^3 universal deformations